

ENGL 119: Communications for Maths and Computer Sciences – Fall Term

Instructor

2026

Patricia Ofili

Email

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Send me an email from Monday to Friday, and I will respond within 48 hours.

**Office Hours: PAS 1284 -
Tue & Thur: 12 – 1:30
pm and by appointment**

Course Description. In this course, you will learn about effective written, oral, and visual communication in mathematics and computer science. You will have the opportunity to shape these communication skills through iterative design processes that emphasize attention to your audience, the purpose of your communications, and student agency. You will work individually and collaboratively to craft messages for internal and external audiences, including scientists, employers, affected communities, or broader publics. You will learn a variety of genres such as research reports, grant proposals, conference papers, public talks, blog posts, and critical analysis of texts. Overall, this course will help you enhance your capacity to conduct research and report findings, communicate effectively/ethically for important changes.

Learning Objectives:

Upon completion of this course, students should be able to:

- design, draft, and persuasively deliver communication to expert and non-expert audiences;
- justify decisions about the language, content, and genre used when communicating;
- practice collaboration and peer review in support of iterative communication design processes, including revision;
- practice research processes to find, assess, document, incorporate, and cite research resources and communicate research findings;
- describe and appraise the purposes and ethical concerns of mathematics and computer science communication.
- In addition, you will acquire confidence in working

as part of a team, communicating, practicing professional behavior, and making oral presentations.

Please note: this is not an English language course. It is strongly recommended that you achieve the English Language Proficiency Exam (ELPE) milestone before taking this course. If English is your second language, you will need to demonstrate a high level of competence in English in order to succeed in this course.

Class Attendance/Engagement and Participation Activities (15%) - Due:

Throughout the term: Class attendance/Participation/Activity is mandatory and worth 15% of your final grade (tallied at the end of term); students must attend every lecture in-person, on time, and demonstrate respectful behavior. A detailed class schedule is attached to this outline. Consult it regularly, as it contains a list of lecture topics, required reading, class activities, and due dates for assignments and other workshop requirements. Active engagement is also expected, which includes listening attentively, participating in discussions (e.g., answering questions when prompted or contributing during small group activities), completing in-class tasks, and using technology only for course-related purposes. Please note that you are required to bring your laptop and writing materials to class, and attendance in all of the class activities/workshops is mandatory.

Required Texts and Materials:

Penrose, Ann M., and Steven B. Katz. *Writing in the Sciences: Exploring Conventions of Scientific Discourse*. 3rd edition. 2020: Open Access Version. (Available online)

Aristotle's "Rhetoric." A summarized version or watch on YouTube: The Art Of Rhetoric: A 30-Minute Summary

Frayling, Christopher: "Curse of the scientist!" Available at ProQuest @uwaterloo Library

T. V. Gerbina, “Science Disinformation: On the Problem of Fake News:” Scientific and Technical Information Processing, 2021, Vol. 48, No. 4, pp. 290–298. © Allerton Press, Inc., 2021.

Tom Carruthers, “Fact-bombing by experts doesn’t change hearts and minds. But good science communication can.” Available at
Academic rigour, journalistic flair

Michael D. Jones¹ & Deseraï Anderson Crow, “How can we use the ‘science of stories’ to produce persuasive scientific stories?” Palmgrave Communications 3, Article No: 35 (2017). Available at <http://www.nature.com/articles/s41599-017>

KJELDTSEN, E, JENS. “What the Metaphor Could Not Tell Us About the Prime Minister’s Bicycle Helmet Rhetorical Criticism of Visual Political Rhetoric.” Available at https://www.academia.edu/2429331/What_the_Metaphor_Could_Not_Tell_Us_About_the_Prime_Minister_s_Bicycle_Helmet_Rhetor

Gloria Snively & John Corsiglia “Indigenous Science: Proven, Practical, and Timeless.” Available at
<https://pressbooks.bccampus.ca/knownhome/chapter/chapter-6/>

Course Assignments and Requirements

Assignment and Evaluation Overview

In this course, you will need to complete the required assignments and activities. In addition, due to the importance of the revision process in writing and communication design, there is no exam:

- **Class Attendance/Participation/Activities 15%**
- **Discussions Posts 10%**
- **Blogposts – 5%**
- **Report: State of Science – 15%**
- **Grant Proposal – 20%**
- **Conference Paper – 20%**
- **Public Communication or Engagement: Group Poster/PowerPoint Presentation – 15%**

Assignment Descriptions

Here is the interesting thing about this course: the assignments will draw upon a range of genres of communication that can be employed in scientific writing. You're going to present your work to a bunch of different audiences. It might be to other scientists, it might be to non-scientists who are just interested in learning about your research, or it might be to people who will give you money to complete your research. We have provided a list of articles and topics that can be used, and we have developed this list in consultation with faculty members in the sciences. If you'd like to choose other texts, you'll have to have them approved.

Contribution Evaluation

Contribution isn't just about posting things on Learn discussion boards. Think of the discussion board posts and engagements as evaluations of some aspects of inter-personal communication. Generally, the expectation is that you treat the discussion board as a professional/virtual classroom space and your peers as your colleagues.

100 points–Exceptional: Frequent, substantive, formative, and original contributions to the discussion board; consistently engaged and participatory comments; outstanding peer feedback

90-99 points–Excellent: Only minor exceptions to the criteria described above

80-89 points–Good: Regular contributor to discussions; consistent engagement; good peer feedback

60-79 points–Satisfactory: Occasional contributions and inconsistent engagement; good peer feedback

50 - 59 points–Marginal: Minimal contributions and/or significant lack of engagement

- 50 points–Failure: Repeated disrespectful engagement with others; consistent lack of commitment and/or effort

Assignments Overview

It's a little early for original research, so we're going to begin with getting you up to speed on engaging with other scientific/non-scientific texts. You will choose an article for your report. You're going to summarize and communicate your report of this article to both expert and public audiences by tearing it apart, critiquing it, identifying a gap in it, and putting your ideas together in an essay. Afterward, you will need to identify a problem in the society, do research on how to resolve that problem, design a grant proposal to help you raise money for your research (virtual research), and come up with your findings. You will write a conference paper that is argumentative in nature to help you persuade your audience on why this problem

you are engaged with is important. To engage with the assignments of Report, Grant Proposal, and Conference Paper, imagine you're a Research Assistant (RA), and these exercises will be the work you're learning about for the term you are hired to work on as an RA.

State of Science Report

Your first major assignment will be a report on the state of science of the article you have chosen. Your goal here is to understand what research is presented in the article you have chosen and critically analyze it. What is the significance of the article in contemporary times?

First, you will want to read and re-read your article. You will need to know the general field, the topic of the article, and why that topic is important. Properly cite your sources to avoid plagiarism using the latest MLA citation format. Draft your report and post it into your group discussion board for peer review and use the peer feedback to edit your work before posting it to the drop box. You should be sure that your assignment is:

- About 4-5 pages in length,
- Times New Roman; double-spaced,
- 12pt font.

Grant Proposal: Crowdfunding Your Research

You will complete this assignment on your own. You will first identify an important problem in the society you would like to find a solution to. You can decide to work on the gap you may have identified in your report. That means you will conduct some research (virtual) on ways that you plan to help solve this problem. Choose an idea that has not been over flogged already, which is relatively new. For this assignment you will create a crowdfunding proposal. Properly cite your sources to avoid plagiarism using the latest MLA citation format. You will submit your draft to the discussion board for peer review. You will use the peer feedback to edit your paper before you submit it to the drop box.

What will you need to complete your project? You will be required to develop a strong argument that justifies your research proposal, why you think it is worth pursuing, and draw up a budget. Include that information and anything else you think is important regarding why you chose the project that you did (e.g., this is a project you are hoping to get funding for). Properly cite your sources to avoid plagiarism using the latest MLA citation format. You should be sure that your assignment is:

- About 4-5 pages in length,

- Times New Roman, double-spaced,
- 12pt font.

Criteria for Evaluation

- Includes all relevant information indicated above.
- Includes all relevant sections of the proposal for your platform of choice (e.g., proposal text, budget, rewards, etc.).
- Clear statement of purpose, following CARS model, and supporting evidence is included.
- The budget should be tailored to your project and audience.
- Technical presentation (typed, well formatted, consistent fonts, etc.).
- Spelling and grammar (including complete sentences)

Conference Paper

Undertake library research to validate your thesis statement. This paper is based on your research findings that prove that the problem you are interested in solving is one that is valid and impactful. Write a 4 – 5-page conference paper that you will post to the discussion board for peer review. Properly cite your sources to avoid plagiarism using the latest MLA citation format. You will use the peer review to edit and refine your paper before the final submission to the drop box. Your paper will be assessed on how clear your ideas are, how logical your arguments are, how clear and sharp your thesis statement is, how well you support your points with credible sources, how strong your introduction and conclusion are, and how well you have edited your work. You should be sure that your assignment is:

- About 4-5 pages in length,
- Times New Roman; double-spaced,
- 12pt font.

Public Communication Group Assignment

This will be a group assignment. Imagine you've been invited to give a talk at your local library about your work. Your audience could be anyone: other scientists, people interested in your topic, students, families looking for an educational opportunity, etc. Demographics include a range of learners, and you're going to have to figure out how to communicate your complex subject to a wide audience. You will first need to decide what aspect of your topic you think will appeal to your audience. What is particularly interesting about your work and what do you most want to share with others? Prepare a slideshow/power point or some other

kind of multimedia to use for showing your group project in the discussion board. Identify a controversial topic in your group, brainstorm on how to address the problem at the center of the controversy, share the major ideas in the topic among yourselves, each person should conduct research about their chosen aspect of the topic and produce slides that utilizes more of visual rhetoric than words for presenting each idea. Each group needs to create physical posters that will be presented/displayed at the General forum on the last day of class.

Criteria for Evaluation

- Clear topic with a narrow focus.
- Well-organized (arrangement!) presentation with clear narrative arc.
- Clear statement of purpose, following CARS model, and supporting evidence is included.
- Creative deployment of visual rhetoric.
- Group cohesion, while showing what individuals worked on.

Course Rules/ Late / Missed Content

Students must complete all course assessments by the specified due dates. Please review the extension policy; extensions apply only to major assignments, are granted solely by the instructor, and must be requested in writing via email. If an extension is approved, the assessment may be submitted by the extended deadline without penalty. All major assignments must be submitted to the LEARN dropbox (no paper or email submissions). In-class participation activities, knowledge checks, rough drafts, and similar components cannot be submitted late because they are interactive or have strict deadlines. Without an approved extension, major assignments lose 2 marks per day if late; submissions more than 3 days late or missing receive a *zero*. Late assignments will have marks docked if you were not granted an extension. To be granted an extension, you must have a compelling reason (e.g., medically documented illness or other related issues). If you know you will need an extension, let me know at least 48 hours prior to the due date. If you are concerned about your grades, contact me within the first 3/4 of the term; the last quarter of the term will not provide sufficient time to markedly improve your final grade. Finally, I will not grant an incomplete in the course; if you have concerns about completing your term please let me know as soon as possible.

Academic Integrity

In order to maintain a culture of academic integrity, members of the University of Waterloo community are expected to promote honesty, trust, fairness, respect and responsibility. See the UWaterloo Academic Integrity webpage and the Arts Academic Integrity webpage for more information.

Discipline

A student is expected to know what constitutes academic integrity, to avoid committing academic offences, and to take responsibility for his/her actions. A student who is unsure whether an action constitutes an offence, or who needs help in learning how to avoid offences (e.g., plagiarism, cheating) or about rules for group work/collaboration should seek guidance from the course professor, academic advisor, or the Undergraduate Associate Dean. When misconduct has been found to have occurred, disciplinary penalties will be imposed under Policy 71 - Student Discipline. For information on categories of offenses and types of penalties, students should refer to Policy 71 - Student Discipline. For typical penalties check Guidelines for the Assessment of Penalties(<https://uwaterloo.ca/secretariat-general-counsel/policies-procedures-guidelines/guidelines/guidelines-assessment-penalties>).

Grievance

A student who believes that a decision affecting some aspect of his/her university life has been unfair or unreasonable may have grounds for initiating a grievance. Read Policy 70 - Student Petitions and Grievances, Section 4 (<https://uwaterloo.ca/secretariat-general-counsel/policies-procedures-guidelines/policy-70>). When in doubt, please be certain to contact the department's administrative assistant who will provide further assistance.

Appeals

A decision made or penalty imposed under Policy 70, Student Petitions and Grievances (other than a petition) or Policy 71, Student Discipline may be appealed if there is a ground. A student who believes he/she has a ground for an appeal should refer to Policy 72, Student Appeals (<https://uwaterloo.ca/secretariat-general-counsel/policies-procedures-guidelines/policy-72>). Note for Students with Disabilities, The AccessAbility Services office, located on the first floor of the Needles Hall extension (NH 1401), collaborates with all academic departments to arrange appropriate accommodations for students with disabilities without compromising the academic integrity of the curriculum. If you require academic accommodations to lessen the impact of your

disability, please register with the AS office at the beginning of each academic term.

Communication for Maths and Computer Sciences

It is your responsibility to email me in advance regarding any confusion you might have about due dates or assignment prompts. Course Schedule is subject to change.

GenAI Policy:

GenAI Use Allowed Under Certain Conditions:

Generative artificial intelligence (GenAI) trained using large language models (LLM) or other methods to produce text, images, music, or code, like Chat GPT, DALL-E, or GitHub CoPilot, may be used in this course under certain conditions with proper documentation, citation, and acknowledgement. Permitted uses of and expectations for using GenAI will be discussed in this course outline/syllabus and also outlined in the major assignment instructions (i.e., prompts). Recommendations for citing GenAI in student work at the University of Waterloo may be found through the library: https://subjectguides.uwaterloo.ca/chatgpt_generative_ai. Please be aware that GenAI is known to falsify references to other work and may fabricate facts and inaccurately express ideas. GenAI generates content based on the input of other human authors and may therefore contain inaccuracies or reflect biases. To protect the privacy and security of any data entered, students should use the University's version of [Copilot](#) and log-in with their UW ID. Data entered into other systems can be added to training sets, monitored, geolocated, and even reproduced as output which may share private personal information or result in intellectual property breaches. In addition, you should be aware that the legal/copyright status of GenAI inputs and outputs is unclear. Exercise caution when using sizable portions of content from AI sources, especially images. More information is available from the Copyright Advisory Committee: <https://uwaterloo.ca/copyright-at-waterloo/teaching/generative-artificial-intelligence>. You are accountable for the content and accuracy of all work that you submit in this class, including any supported by GenAI.

Below are the GenAI use conditions (i.e., the parameters) set by the instructor:

What Is Allowed:

In this course, *students may use GenAI tools (e.g., Copilot) for the purposes of researching and 'revising' the three major assignments only, as specified below.* To avoid academic misconduct, you must properly *document and cite all GenAI use*; for examples of proper GenAI documentation and citation, see this

link: <https://uwaterloo.ca/writing-and-communication-centre/Resources-AI-Citing> (check out the ‘example documentation table’ and ‘Microsoft Word Template’).

You may refine or expand ideas the student has already generated; clarify research questions; organize those ideas into a basic outline or rough draft of no more than 200 words created from the student’s own ideas; generate keywords; identify possible examples or credible sources to investigate; and format citations in APA or MLA style; all while ensuring that students themselves create all ideas (e.g., arguments, analyses, etc.) for the assignment. See link:

link: <https://uwaterloo.ca/writing-and-communication-centre/Resources-AI-Drafting>

‘Revising:’ Suggest small-scale edits to a student-written draft regarding clarity, sentence-level expression, logical flow, grammar, and organization; point out gaps or areas needing stronger support; and offer limited alternative phrasings; all while preserving the student’s meaning, without rewriting full paragraphs or supplying new arguments, evidence, or interpretations. See link: <https://uwaterloo.ca/writing-and-communication-centre/Resources-AI-Revising>

Rationale for the Use Limits: These ‘drafting’ and ‘revising’ limits ensure that GenAI is used only to support planning, clarity, and organization, not to replace the student’s own ideas, writing voice, etc. GenAI literacy skills may thus be incorporated into the course (through early-stage scaffolding and small-scale edits), but the student remains responsible for the core intellectual work of each major assignment.

What Is Not Allowed:

Students may not use GenAI tools (e.g., Copilot) to generate their ideas (e.g., arguments, analyses, etc) or to write or rewrite substantial portions of their major assignments (i.e., more than 200 words). In essence, GenAI should not replace the student’s own critical thinking, expression, or writing voice, and therefore may not be used to produce the good copies of the three major assignments or to complete any of the in-class participation activities, knowledge checks, basic outlines or rough drafts beyond the first one, or peer review comments. Students also may not input assignment prompts, rubrics, or any other instructor-, TA-, or peer-created materials into GenAI platforms due to intellectual property and privacy concerns. All good copies must be written primarily in the student’s own words and reflect their own original ideas (except where properly paraphrasing/quoting and citing credible, outside sources). Unauthorized use in this course (i.e., beyond ‘drafting’ and ‘revising’ the three major assignments only, as specified above) will be considered in violation of the fundamental values of academic integrity, including

honesty, trust, respect, fairness, responsibility, and courage ([ICAI](#), n.d.) and subject to [Policy 71—Student Discipline](#) (plagiarism or unauthorized aids or assistance). To avoid academic misconduct, students should be prepared to demonstrate their own learning, thought, and writing processes; *students will be asked to submit evidence of their rough work*, either through saved drafts or document versions, along with the good copies of their major assignments. If the use of GenAI is suspected where not permitted, then the student may be asked to meet with the instructor to provide explanations to support the submitted final material as being their original work. Through this process, if the student has not sufficiently supported their final submissions (i.e., good copies), academic misconduct allegations might then be brought to the Associate Dean. Students are encouraged to reach out to the instructor via email well before the major assignment due dates if they have any questions about what is/not allowed in terms of GenAI tool use. Students are also encouraged to access campus supports if they need help with their coursework, including the:

- [Student Success Office](#) for help with skills like note taking and time management.
- [Writing and Communication Centre](#) for assignments with writing or presentations.
- [AccessAbility Services](#) for documented accommodations.
- [UW Library](#) for research-based assignments.
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Teaching and Learning with AI:

GenAI tools (i.e., “Microsoft Copilot used through your UWaterloo account to protect the privacy and security of the data you enter”) may be used by the instructor/TA in this course *in ways that follow the University of Waterloo’s official guidance on teaching and learning with AI*. In this course, that might include using GenAI to generate short practice materials (e.g., prompts, sample texts, or worksheet style exercises) to support students’ skill development. GenAI may also be used for limited revision support, such as checking grammar or clarity in instructor- or TA-written text. The University’s guidance, provided in the citation and quoted excerpt below, informs these instructional uses:

Citation:

University of Waterloo. (2024). *Artificial Intelligence at Waterloo*. Associate Vice-President, Academic. <https://uwaterloo.ca/associate-vice-president-academic/artificial-intelligence-uw>.

Administrative Policy
Mental Health Support

All of us need a support system. The faculty and staff in Arts encourage students to seek out mental health support if they are needed.

On campus supports

For counselling (individual or group) reach out to [Campus Wellness and Counselling Services](#). Counselling Services strives to provide a secure, supportive environment for students of all orientations and backgrounds. They offer confidential counselling for a variety of areas including anxiety, stress management, depression, grief, substance use, sexuality, relationship issues, and much more.

Other on-campus supports

- [MATES](#): one-to-one peer support program offered by the Waterloo Undergraduate Student Association (WUSA) and Counselling Services
- Download the [Regroup Mobile](#) to your phone to quickly access mental health support information
- [Empower Me](#): to access create an account on the Dialogue mobile app or on the web at www.studentcare.ca/dialogue
- [Sexual Violence Prevention and Response Centre](#) - supports all members of the University of Waterloo campus community who have experienced or been impacted by sexual violence.

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Off campus supports

- [Good2Talk](#): Free confidential help line for post-secondary students. Phone: 1-866-925-5454
- Grand River Hospital: Emergency care for mental health crisis. Phone: 519-749-4300
- [St. Mary's Hospital](#) - 519-744-3311
- [Here 24/7](#): Mental Health and Crisis Service Team. Phone: 1-844-437-3247
- 988 Suicide Crisis Helpline: a 24/7 urgent mental health care three-digit helpline
- [OK2BME](#): set of support services for Two-Spirit, lesbian, gay, bisexual, trans, queer, intersex, asexual (@SLGBTQIA+) and questioning kids, teens, adults, and their families in Waterloo Region. Phone: 519-884-0000 or e-mail ok2bme@caminowellbeing.ca
- [Sexual Assault Support Centre of Waterloo Region](#) - offers individual counselling to people of all genders, 16+ who have been sexually assaulted at any point in their lives. As well as short-term support for family members or friends of sexual assault survivors. Also, information and referrals

More information about resources and supports for students can be found online in on the [Faculty of Arts Student Support](#) page.

Absences due to verified illness

For absences due to illness that are not pandemic-related, students need to obtain a [Verification of Illness or Extenuating Circumstances Process](#). Submission through the [Absence Verification System](#), once registered, will notify your instructors of your absence. Students should not submit their VIF or any other medical documentation to your instructors, teaching assistants, or other course personnel. To request and arrange accommodations, it is the student's responsibility to contact the instructor within 48 hours of the date of any missed course component, or as soon as the student can, given their health condition.

Absences due to extenuating circumstances (e.g., bereavement)

For absences due to extenuating circumstances (e.g., bereavement, serious family illness), students submit [a request for absence due to extenuating circumstances](#) to the [Absence Verification System](#) for processing. Students use the form to submit supporting documentation, which will be considered in a comparable manner to a VIF. Official documentation is necessary before any action can be taken. Examples are plane/train/bus tickets, court documents, police reports, death certificates/obituaries/letters from funeral directors. To request and arrange accommodations, it is the student's responsibility to contact the instructor within 48 hours of the date of any missed course component, or as soon as the student can, given their circumstances.

Self-declared short-term absences for any reason

Students can [self-declare](#) one short-term absence per term on Quest for any reason. A short-term absence covers two calendar days, and no documentation is required. The student has only one self-declared short-term absence per term and this absence can only be used during the formal lecture period. Submitting the self-declaration through Quest will notify your instructors of your absence. It is the student's responsibility to contact their instructors within the first 24 hours after submitting their self-declaration to discuss accommodations for missed course components.

Accommodating religious and spiritual observances

The University of Waterloo has a duty to [accommodate religious, spiritual, and other creed-based beliefs and practices](#) under the Ontario Human Rights Commission (2015) [Policy on preventing discrimination based on creed](#). Students may seek accommodations for missed course components on religious, spiritual, or

other creed grounds. In such cases the students should please consult the instructor within two weeks of the announcement of the due date for which the accommodation is being sought. Students also may request accommodations for temporary absences from classes or other course-related activities to engage in prayer or other daily spiritual practices. Instructors are expected to make reasonable arrangements to respectfully accommodate such requests. The University of Waterloo has a number of [multi-faith spaces](#) that students can use for faith-based practices.

Date	Topic, Material and Readings.	Due Dates and Description of assignments
<i>Sept 10</i>	Communication/Introduction/Syllabus Homework: Read Aristotle’s “Rhetoric.” A summarized version or watch on YouTube: The Art Of Rhetoric: A 30-Minute Summary & Frayling, Christopher: “Curse of the scientist!”	What is Ethos, Pathos, and Logos, and how do they operate in science communication through various mediums of communication like Facebook, Instagram, Twitter, and other mediums of communication? Submit your short analysis of Aristotle’s concepts to the Discussion Board (250 words). Due on Sept 14: Submit your short analysis of Aristotle’s concepts to the Discussion Board (250 words). To be discussed in class the next day.
<i>Sept 15</i>	Class Activities: Discuss Aristotle’s three modes of persuasion & Frayling, Christopher: “Curse of the scientist!” In the first week of classes, we will	Due – Sept 16: Post your analysis of “Science Disinformation: On the Problem of Fake News” by T. V. Gerbina on the Discussion Board to be discussed in class

	discuss the idea of genre in rhetorical studies. Students will be introduced to this concept through a series of examples of genres in science communication. Homework: Read – “Science Disinformation: On the Problem of Fake News” by T. V. Gerbina And show how Ethos, Pathos, and Logos function in the article	the next day.
<i>Sept 17</i>	Discuss – T. V. Gerbina arguments Discuss how T. V. Gerbina deployed ethos, pathos, and logos to discuss the concept of disinformation. Argumentation–Getting your evidence in order. Arrangement is one of the core concepts in rhetoric, and this idea builds on our discussion of genre. Homework: Read: (pages 1 – 10) KJELDSSEN, E, JENS. “What the Metaphor Could Not Tell Us About the Prime Minister’s Bicycle Helmet Rhetorical Criticism of Visual Political Rhetoric.” Available at https://www.academia.edu/2429331/What_the_Metaphor_Could_Not_Tell_Us_About_the_Prime_Minister_s_Bicycle_Helmet_Rhetor	Due on Sept 18: Post your ideas about Visual Rhetoric in KJELDSSEN, E, JENS’s article (250 words) on the discussion board to be discussed in class the next day.

<i>Sept 22</i>	Discuss your ideas about Visual Rhetoric in KJELDTSEN, E, JENS's article What is Visual Rhetoric? Investigate the effectiveness of Visual rhetoric. Visuals and visualization are different modes of communication that are crucial to scientific research. We will explore the different ways that visuals can operate across genres of science communication. As well, we will explore issues of data visualization and ``anticipate questions of ethical concerns in addition to readability and clarity.	
<i>Sept 24</i>	Library Workshop This workshop is recommended for everyone, and it will help you learn more about how to do productive research that is absolutely necessary for your entire academic pursuit. Class Activity – Discuss and write a short note on: “Technology and the Visual Dimension of Science” (pg 32) Homework: Read - Penrose, Ann M. and Steven B. Katz's “Technology and the Visual Dimension of Science” (pg 32) & Do a blogpost by making a persuasive	Due Sept 25 Library Workshop – Post your experience of the Library Workshop on Learn Discussion Board What makes a good blogpost? How do Scientific Visuals and Visualization enhance persuasion? With “Technology and the Visual Dimension of Science” in mind - make a Blogpost by making a persuasive argument about your favorite gadgets, app, or a controversial phenomenon like climate change, AI proliferation, etc. Look for a Peer Review article to

	argument about your favorite gadgets, app, or a controversial phenomenon like climate change, AI proliferation, etc (use good visual rhetoric) and post it on the discussion board on Learn. This blogpost counts towards 5% of your grade.	analyze for your State of Science Report On Sept 28, Submit your Blogpost on the Discussion Board, to be discussed in class on Sept 29
Sept 29	Class Activity – “Technology and the Visual Dimension of Science” (pg 32) & Evaluate individual blogposts, particularly the visual rhetoric deployed in the posts Homework: Draft your State of Science Report & Read Tom Carruthers, “Fact-bombing by experts doesn’t change hearts and minds. But good science communication can.”	Draft our State of Science Report on any peer review article of your choice
Oct 1	Class Activity – Discuss Tom Carruthers, “Fact-bombing by experts doesn’t change hearts and minds. But good science communication can.”) & Writing Strategies Homework - Read- “ SCIENTIFIC COMMUNICATION AND CONVENTION” (Pg. 14) &	On Oct 5, Post the draft of your report on the discussion board for peer review- give a review of three of your peers’ essays. Your peer review counts towards participation grade.

	Draft your Report	
Oct 6	Class Activity - Peer Review Workshop/Peer Review of the State of Science Report “SCIENTIFIC COMMUNICATION AND CONVENTION” (Pg. 14) Give feedback on the Reports of your peers Homework: Revise your draft Homework: Read – “TECHNOLOGY AND THE RHETORICAL CHALLENGE OF COMMUNICATING SCIENCE” (pg 41)	Use the feedback you received to revise your essay. Review your colleague’s report, build on the previous discussions of arrangement and argument. On Oct 7, Submit your report as word attachment into the drop box.
Oct 8	Class Activity – Discuss and write a short note on: - “TECHNOLOGY AND THE RHETORICAL CHALLENGE OF COMMUNICATING SCIENCE” (pg 41) Homework: Watch on YouTube: Sharing science through story: Fergus McAuliffe at TEDxDublin Read: Michael D. Jones¹ & Desera Anderson Crow, “‘How can we use the ‘science of	On Oct 9, post your analysis of “TECHNOLOGY AND THE RHETORICAL CHALLENGE OF COMMUNICATING SCIENCE” (pg 41) in (250 words) on the Discussion Board As scientists, describe as effectively as possible how ordinary people can use technology efficiently, navigate social media platforms, and consume the information presented in a critical manner.

	stories’ to produce persuasive scientific stories?” Homework: Read – “COMMUNICATION ETHICS IN SOCIAL CONTEXTS ” (44) “Introducing the Research Problem” (79)	
Oct 13	Reading Week	
Oct 15	Reading Week	
Oct 20	Class Activities- Discuss: how science communication is changing in Fergus McAuliffe’s Ted Talk and Michael D. Jones1 & Deserai Anderson Crow, “How can we use the ‘science of stories’ to produce persuasive scientific stories?” & Why is Grammar Important in Writing? Class Activity – Discuss and write a short note on: “COMMUNICATION ETHICS IN SOCIAL CONTEXTS ” (44) “Introducing the Research Problem” (79) Start your next major assignment: Grant Proposal. Next assignment: Grant Proposal-	Due on Oct 21: Post your analysis of Fergus McAuliffe’s Ted Talk and Michael D. Jones1 & Deserai Anderson Crow article on the discussion board (250 words). Think of creative ways through which you can get funding for your brainchild. Identify a gap in any research area or modern Tech/Artificial Intelligence space you are interested in and conduct the research for your grant proposal. Establish a territory for your research, a gap in the research, and then explain how you’re going to fill the gap.

	Identify a problem or pick an idea (by identifying a gap you want to fill) from the article chosen for your report. Homework: Prepare your grant proposal Increasingly, research doesn't happen without securing external funds. This research project explores the complex task of writing a grant application. Homework: Read – Create a Research Space – (pg 81) “Describing Methods” (82) “Reporting Results” (86)	
Oct 22	Class Activity – Elements of a Grant Proposal Discuss and write a short note on: Create a Research Space – (pg 81) “Describing Methods” (82) “Reporting Results” (86) Discuss - your Grant Proposal Topics. Homework –Read – “ETHICAL “COSTS” OF COMMUNICATION TECHNOLOGY” (Pg. 58)	How do your findings fill the gap you have identified?

Oct 27	Class Activity – Discuss and write a short note on: Discuss and write a short note on: “ETHICAL “COSTS” OF COMMUNICATION TECHNOLOGY” (Pg. 58) Discuss: How science communication is changing	On Oct 28, post your Grant Proposal draft on the discussion board for peer review. Give peer reviews of three of your peers’ essay. Your peer review counts towards participation grade. Use the feedback to revise your essay.
Oct 29	Class Activity – Peer Review of Grant Proposal Draft: Use the feedback to revise your essay. Homework: Read “THE ROLE OF RESEARCH CONFERENCES IN THE SCIENCES” (123) “WRITING CONFERENCE PROPOSAL ABSTRACTS” (124)	On Oct 30, Submit your grant proposal as word attachment into the drop box.
Nov 3	Class Activity – Discuss and write a short note on: “THE ROLE OF RESEARCH CONFERENCES IN THE SCIENCES” (123)	Public Communication of sciences and its impact on public policies. EDGE workshop – this workshop is recommended to enable you translate

	“WRITING CONFERENCE PROPOSAL ABSTRACTS” (124) Homework: Read “Organizing the Research Talk” 127 “ Methods of Oral Presentation” (129)	your skills into your resumes, interviews, and the workplace.
Nov 5	Effective Communication: How Your Communication Skills Work Class Activity – Discuss and write a short note on: “Organizing the Research Talk” 127 “ Methods of Oral Presentation” (129) Homework: Read: Gloria Snively & John Corsiglia “Indigenous Science: Proven, Practical, and Timeless.”	
Nov 10	Class Activity – Discuss: Gloria Snively & John Corsiglia Ethics/Ethos in Science/Science Communication Writing a Conference Paper Discuss and Draft your conference paper to be submitted to the discussion board.	Due on Nov 11: Post your analysis of Gloria Snively & John Corsiglia - (250 words) to the Discussion Board

	Homework: Read “Scientific Communication and Public Communication: An Ethical Conflict?” (62)	
Nov 12	Class Activity – Discuss and write a short note on: “Scientific Communication and Public Communication: An Ethical Conflict?” (62) Peer Review of Conference Paper Use the feedback to revise your essay. Homework: Read “The Role of Collaboration in Scientific Communication” (16) “Technology and Collaboration in Science” (22)	On Nov 11, Post your conference paper draft on the discussion board for peer review. On Nov 13, Submit your conference paper as word attachment into the drop box.
Nov 17	Class Activity – Discuss and write a short note on: “The Role of Collaboration in Scientific Communication” (16) “Technology and Collaboration in Science” (22) Groups and Collaboration in the Sciences	I have created group forums on Learn Discussion board for the group project. Brainstorm and identify a controversial topic that is currently a problem the society is dealing with and to which you need to find a solution (Avoid topics with religious undertones or sensitive subject matter). The topic should be controversial and of importance to society in contemporary times.

	Discuss the group projects you have been involved with in the past. Homework: Read “Understanding “General” Audiences” (167)	
Nov 19	Class Activity – Discuss and write a short note on: “Understanding “General” Audiences” (167) Public Speaking/Communication and the Sciences Homework: Read “ Adapting Through Narration” (171) “ Adapting Through Examples” (172) “ Adapting Through Graphics” (176)	Group Project: The group will put together slides that will be posted on the discussion board for everyone to see. Discuss among yourselves on how you plan to put your group presentations and slides together.
Nov 24	Class Activity – Discuss and write a short note on: “ Adapting Through Narration” (171) “ Adapting Through Examples” (172) “ Adapting Through Graphics” (176) Intercultural Communication Final project: Discuss the topic for the group project.	Prepare your slides and posters for presentation in class.

	Homework: Read “Methods of Oral Presentation” (129) “The Use of Graphics in Oral Presentations” (133)	
Nov 26	Class Activity – Discuss and write a short note on: “Methods of Oral Presentation” (129) “The Use of Graphics in Oral Presentations” (133) Presentation Tips	Work on your group project Finalize your slides and posters for presentation in class.
Dec 1	Group Presentation	Group Assessment
Dec 3	Group Presentation	Group Assessment
Dec 8	Last Class – General Forum/ Group Poster Presentation Class Period @ STC 2002 (2nd floor foyer above and east of the STC Starbucks)	Submit your Power Point Slides and a picture of your group poster to the Discussion Board.